

CHAPTER:8

PUMPS FOR GROUNDWATER LIFTING

Topic of Presentation

Jet Pump:
Working Principle and Suitability

Subject: Groundwater Engineering

Tutor: Ms. Sujata Adhikari

Introduction to Jet Pump

- 1) Jet pumps are fluid entrainment devices that transfer energy from a high speed motive fluid to a suction fluid without moving mechanical parts, using Venturi principles to create suction and lift fluids.
- 2) It operates by forcing a "motive" fluid through a nozzle to create a pressure drop that draws in groundwater.
- 3) Unlike standard mechanical pumps, the pumping action is achieved without any moving parts inside the well itself.
- 4) This device is particularly effective in specialized geotechnical scenarios where traditional pumping equipment might fail.

Introduction to Jet Pump

- The system relies on the interaction between a nozzle, throat, and diffuser to convert velocity energy into pressure energy.
- Jet pumps are also referred to as ejector pumps in groundwater engineering literature

Venturi Effect:

The Venturi effect is a fluid-flow phenomenon in which the pressure of a fluid decreases when it flows through a constricted section of a pipe while its velocity increases.

Components of Jet Pump

The jet pump consists of four critical parts:

- **Nozzle:** Converts the pressure energy of the power fluid into high velocity kinetic energy.
- **Suction Chamber:** The area where the vacuum is formed and the secondary fluid enters.
- **Mixing Tube (Throat):** The cylindrical section where the two fluids exchange momentum and reach a uniform velocity.
- **Diffuser:** A divergent section that recovers pressure before discharge.

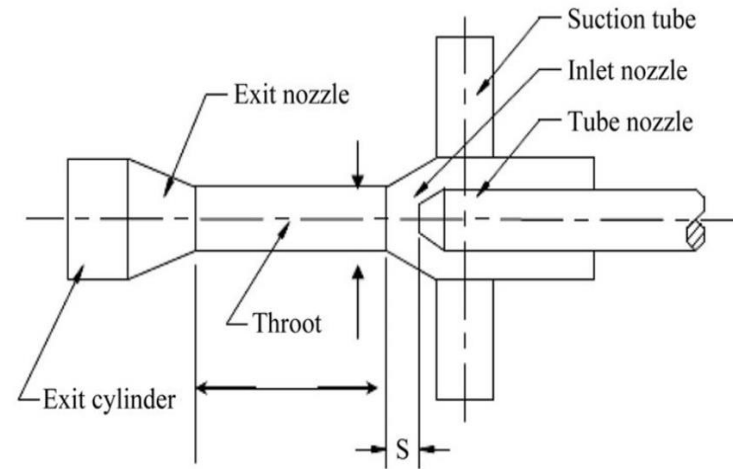


Fig: Jet Pump

Components of Jet Pump

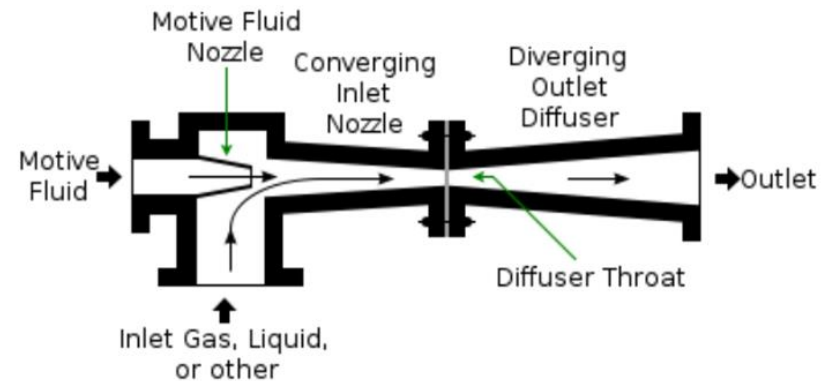
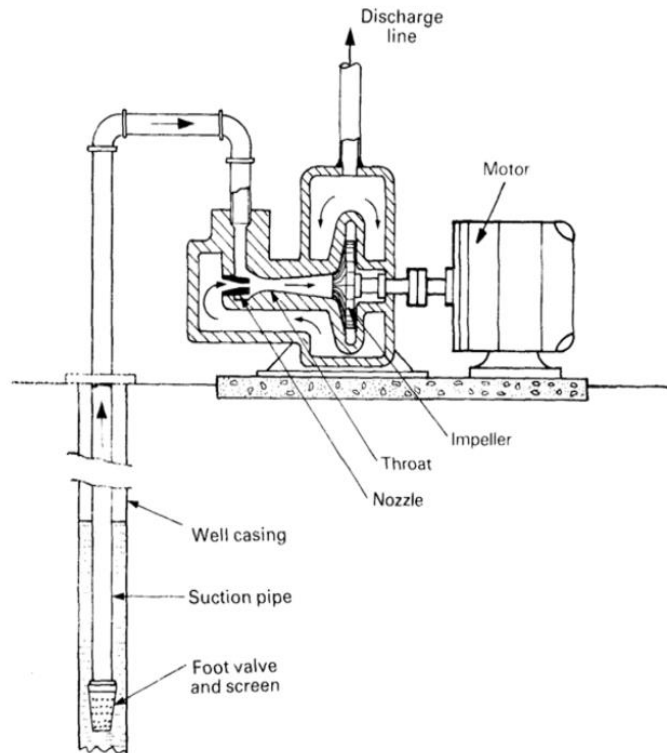


Fig: Jet Pump

Working Principle of Jet Pump

The jet pump operates primarily on the **Venturi Effect** and the principle of **momentum transfer**.

- **Energy Conversion:** A high-pressure "motive" or "power" fluid (liquid or gas) is forced through a constricted nozzle. As the fluid accelerates, its static pressure drops significantly, creating a partial vacuum
- **Suction and Mixing:** This low-pressure zone draws in a secondary "suction" fluid from the reservoir or source. The two fluids enter a mixing tube where the high-velocity motive fluid transfers momentum to the lower velocity suction fluid
- **Pressure Recovery:** The combined mixture then enters a divergent diffuser. Here, the velocity decreases, and the kinetic energy is converted back into static pressure, allowing the fluid to be discharged at a pressure higher than the suction pressure

The Working Mechanism

- Step 1: High Pressure motive fluid enters the nozzle
- Step 2: Velocity increase $\rightarrow$ Pressure drops below atmospheric pressure(Bernoulli's Principle)
- Step 3: Secondary fluid is “sucked” into the low pressure zone.
- Step 4: Both fluids mix in the throat, equalizing velocity.
- Step 5: The diffuser slows the flow, boosting discharge pressure.

Types of Jet Pump

SINGLE- PIPE JET PUMP:

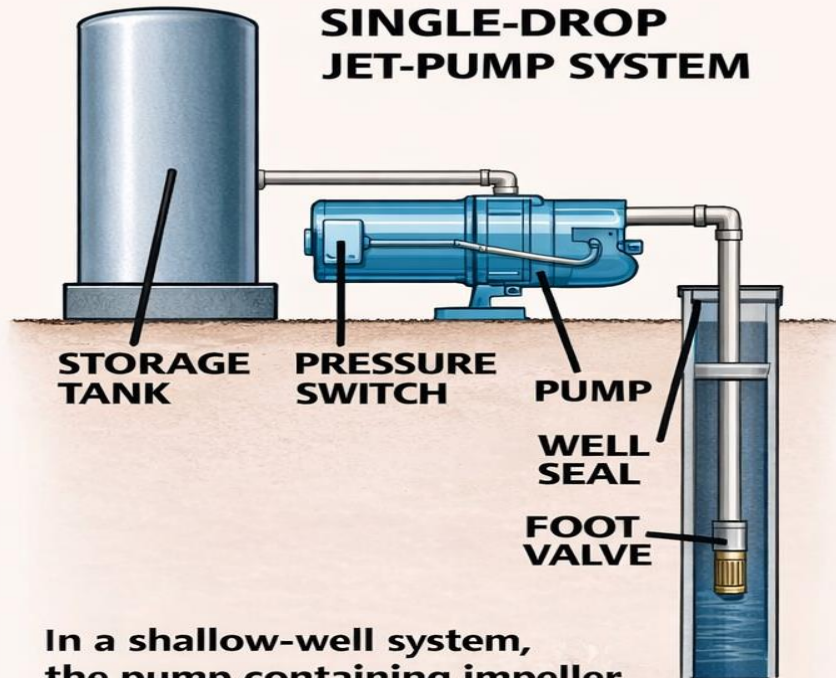
- Best for shallow wells with water levels up to 25 feet deep.
- Uses on suction pipe to draw water up. The jet assembly(venturi and nozzle) is built directly into the pump housing above the ground.
- It relies primarily on suction created by the pump to pull water to the surface.

Double – Pipe Jet Pumps:

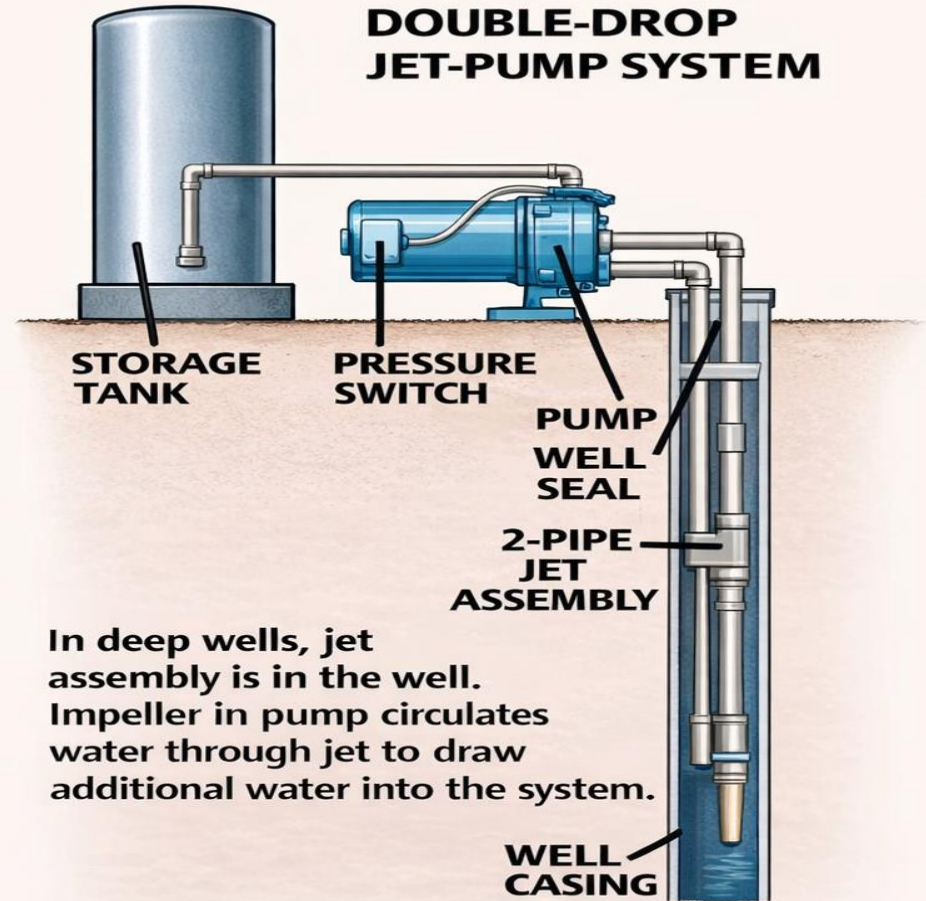
- Designed for deeper wells, typically 25-110 feet.
- Uses two pipes running into the well: Pressure pipe pushes water down into the well and the suction (return) pipe pulls water back up to the surface.
- The jet assembly is placed down in the well rather than in the pump. Water is forced down the pressure pipe through the submerged jet to create a vacuum, which helps “push” more water back up the large suction pipe.

The choice depends strictly on the vertical distance to the water table and the required discharge head.

Types of Jet Pump



In a shallow-well system, the pump containing impeller and jet assembly is aboveground. Atmospheric pressure limits well depth to about 25 ft.



In deep wells, jet assembly is in the well. Impeller in pump circulates water through jet to draw additional water into the system.

Suitability of Jet Pump

- **Abrasive Environments:** Because there are no submerged moving parts, jet pumps are exceptionally suited for wells containing sand, silt, or muddy water that would otherwise erode mechanical seals in submersible pumps.
- **Small Diameter Boreholes:** Their slim profile allows them to be installed in very narrow casings where other pumps might not fit.
- **Corrosive Conditions:** The jet assembly can be manufactured from non-corrosive materials (like plastic or bronze), making them ideal for pumping contaminated or chemically aggressive groundwater
- **Ease of Maintenance:** The mechanical motor and electrical components remain at the surface, allowing for easy access and repair without pulling the entire pipe string.
- Can operate reliably in areas with fluctuating groundwater levels.
- **Self-Priming Capability:** Can clear air from suction lines automatically.

Applications of Jet Pump

- **Groundwater extraction** from deep or narrow wells.
- **Aerospace:** Fuel pumping and cooling systems in high vibration environments
- **Oil Wells:** Used in "Artificial Lift" for high-viscosity or sandy crude.
- **Subsea Mining:** Annular Jet Pumps (AJPs) transport slurries from the ocean floor.
- **Water Treatment:** Aeration and chemical dosing.
- **Temporary pumping installations** in construction projects

Chemical dosing is the controlled addition of specific chemicals into a system (water, wastewater, industrial process, soil, or fluid stream) in accurately measured amounts to achieve a desired chemical, physical, or biological effect.

Limitations of Jet Pump

- **Low Efficiency:** Lower than centrifugal pumps due to mixing losses.
- **High Motive Demand:** Requires a continuous supply of high-pressure fluid.
- **Depth Sensitive:** Performance drops as suction lift increases beyond certain limits.
- Not suitable for very high-yield aquifers
- **Power Requirements:** They require a continuous supply of pressurized motive fluid (or compressed air in similar air-lift systems) to function.

Conclusion:

- Jet pumps offer a robust solution for groundwater extraction in narrow or sandy boreholes.
- The absence of submerged moving parts ensures long-term durability and lower maintenance costs.
- While they have lower hydraulic efficiency, their ability to handle "dirty" water makes them indispensable in specific engineering projects.

References:

- Gosline, J. E., & O'Brien, M. P. (1933). The Water Jet Pump. *University of California Publications in Engineering*.
- Karassik, I. J., et al. (2008). *Pump Handbook*. McGraw-Hill Education.
- Liang, A. (2014). Research on Status and Development of Jet Pump. *Atlantis Press*.
- White, F. M. (2016). *Fluid Mechanics*. McGraw-Hill.
- Abou Taleb, S., & Al-jarrah, A. M. (2017). Performance of airlift and jet pumps. *Journal of Fluid Mechanics*
- El-Dessouky, H., et al. (2002). Performance analysis of water jet pumps. Desalination
- GlobalSpec. (n.d.). *Jet pumps: Selection and information*.
https://www.globalspec.com/learnmore/flow_transfer_control/pumps/jet_pumps
- Driscoll, F. G. (1986). Groundwater and Wells.



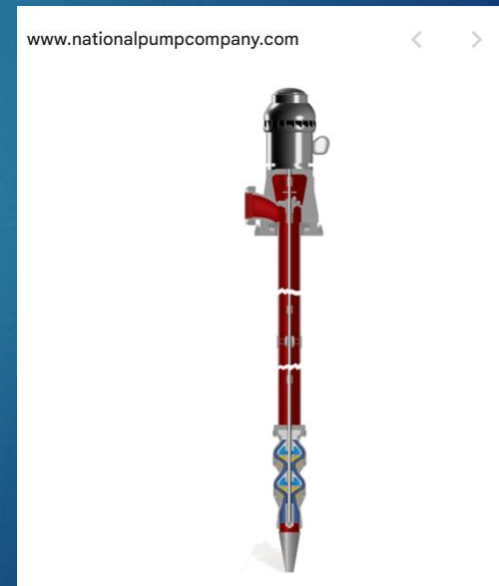
Deep-Well Vertical Turbine Pump

WORKING PRINCIPLE AND SUITABILITY

Nishan K.C.
078bce098
Groundwater
Engineering

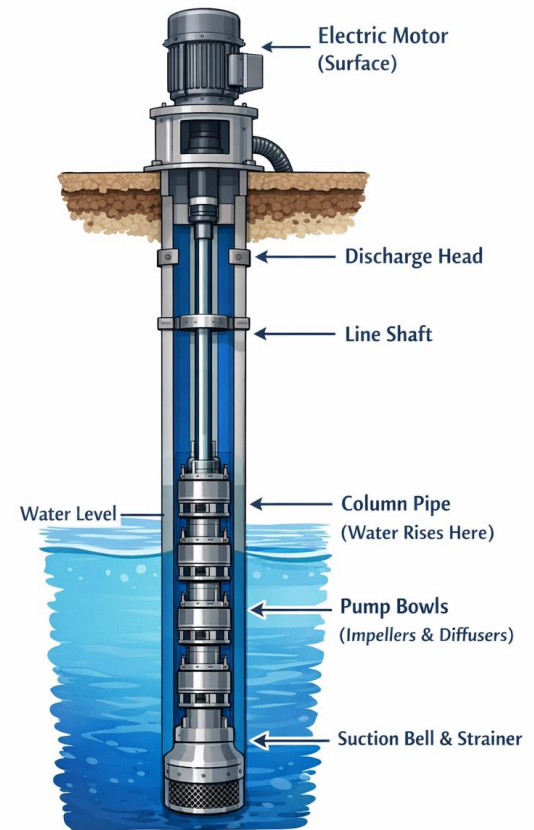
Introduction

- ▶ Deep-Well Vertical Turbine Pump (DWVTP) is a vertical pump designed to lift water from deep wells, reservoirs, or rivers
- ▶ Commonly used where water level is far below ground surface
- ▶ Widely applied in irrigation, municipal and industrial water supply



Main Components

Part	Main role
Motor	Provides power
Line shaft	Transmits power
Column pipe	Carries water upward
Pump bowls	Increase pressure
Suction bell	Allows water entry



Deep-Well Vertical Turbine Pump

Working Principle

Motor rotates the vertical shaft

Each impeller adds energy to the water

Pressure increases stage by stage and water flows upward

Water is delivered to the surface through column pipe

Suitability of Vertical Turbine Pump

Suitable for deep wells where suction lift is not possible

Best for large discharge and high head requirements

Applicable for continuous and long-term operation

Performs well in clean groundwater conditions

Advantages

- ▶ High efficiency for deep wells
- ▶ Motor remains at ground level for easy maintenance
- ▶ Can handle large discharge
- ▶ Long service life

Limitations

- ▶ High initial and installation cost
- ▶ Requires skilled installation and alignment
- ▶ Not suitable for small discharge systems
- ▶ Performance affected by sandy or corrosive water

Applications

- ▶ Deep tube wells
- ▶ Irrigation projects
- ▶ Municipal water supply systems
- ▶ Industrial water extraction

Conclusion

- ▶ Deep-well vertical turbine pump is ideal for deep groundwater lifting
- ▶ Efficient and reliable for large-scale water supply
- ▶ Proper design and maintenance ensure long service life



THANKYOU.

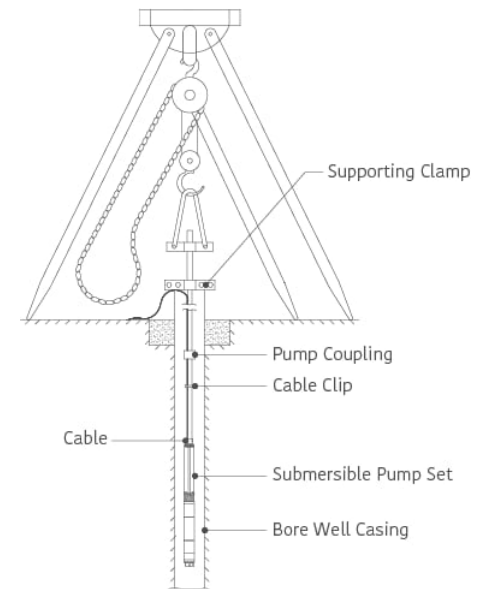
Submersible Pump

SUBMITTED BY : PRAMOD CHAULAGAIAN

078BCE109

1. Introduction

A submersible pump is a centrifugal pump in which the pump and its electric motor are completely submerged in the liquid to be pumped, and it works by converting mechanical energy into hydraulic energy using centrifugal force to lift the fluid to the required height

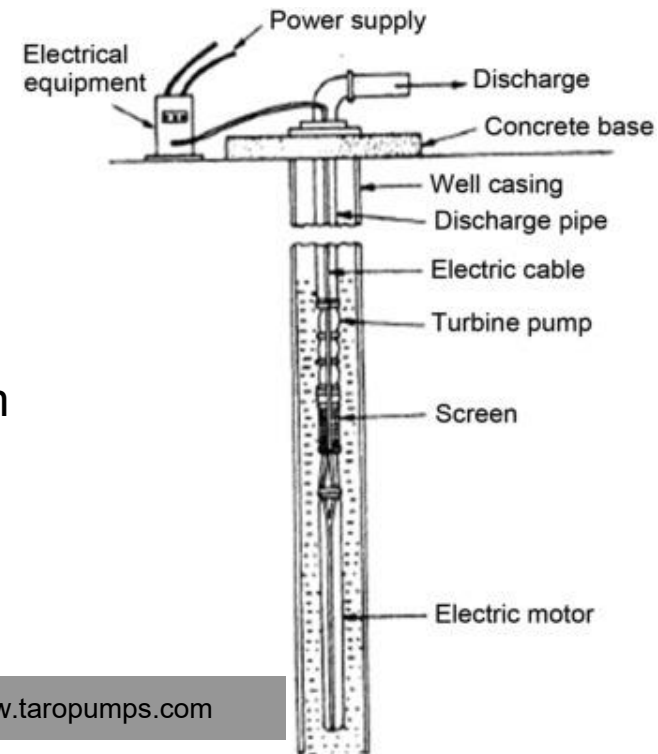


<https://www.taropumps.com>

2. Working Principle

Basic Principle

- Operates on the centrifugal force principle.
- When the motor rotates the impeller, water gains velocity and pressure.
- Water is forced toward the outlet as it moves from high-speed to high-pressure zones



3. Main Components

- ❑ Electric Motor – Provides mechanical energy.
- ❑ Impeller – Creates centrifugal force.
- ❑ Pump Casing – Houses impeller and guides flow.
- ❑ Seal & Shaft – Prevents leakage.
- ❑ Discharge Outlet – Directs water to delivery system

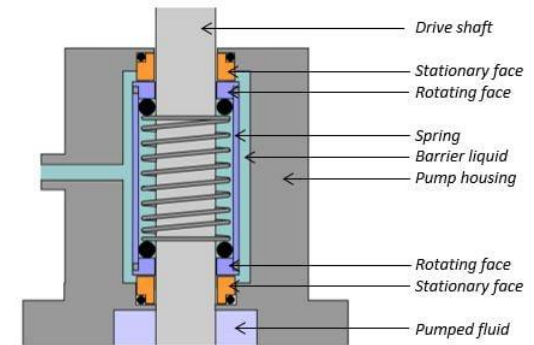


Figure 4. Double mechanical seal

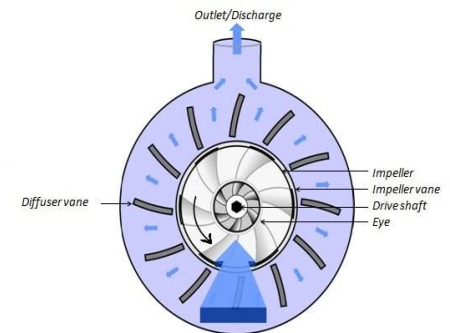


Figure 3. Diffuser case design

4. Step by step working

☐ **Motor Rotation:**

An electric motor drives the impeller.

☐ **Fluid Entry:**

Fluid enters the pump through inlet openings at the bottom.

☐ **Centrifugal Action:**

The impeller accelerates the fluid outward, creating pressure.

☐ **Outlet Discharge:**

Pressurized fluid moves up through the discharge pipe to the surface

5. Advantages

- Operates while fully submerged
- Reduces cavitation losses
- Quiet operation
- High efficiency in deep installations
- Suitable for high lift application

6. Suitability

Submersible pumps are suitable for:

- ☐ Deep well water supply
- ☐ Groundwater extraction
- ☐ Irrigation systems
- ☐ Industrial fluid transfer
- ☐ Drainage and sewage pumping

7. why suitable

- ☐ Works in deep water levels
- ☐ Avoids cavitation
- ☐ Higher pressure head capabilities
- ☐ Reduced maintenance cost

8. Summary

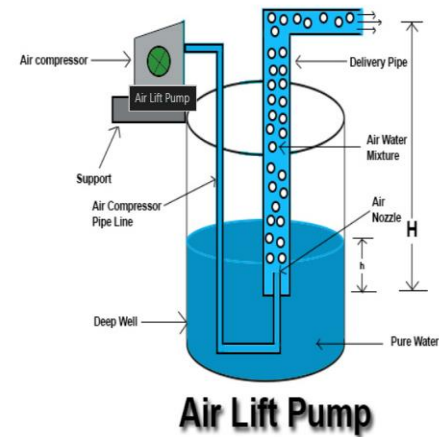
Principle	=	Centrifugal action in submerged conditions
Main Use	=	Water supply, irrigation, drainage
Best for	=	Deep wells & high lift requirements
Advantages	=	High efficiency, low noise, no cavitation

Air lift Pump

Presented by
Ronish Sapkota
078BCE128

Teacher
Ms. Sujata Adhikari

An airlift pump is a device used to raise the liquid from a well or vessels through a vertical pipe, partially submerged in liquid by means of compressed air introduced into the pipe from the lower end (Abou Taleb & Al-jarrah, 2017).



Overview of Airlift Pumps

- Air lift pumps are used in an extensive range of applications because of their simple construction, cost-effective maintenance, easy performance, and high reliability(Mahrous, 2013).
- Air-lift pumps are used in specialized applications, where more conventional pumps cannot operate.
- It can be used for pumping hydrocarbons, dirty water, dangerous liquids, and viscous liquids(Di Gerlando et al., 2010), biochemical applications in bioreactors (Zimmerman et al., 2009), and for seabed mining (Pougatch & Salcudean, 2008).

Suitability of airlift pump

- Deep wells and boreholes
- Sandy, muddy, and abrasive water
- Corrosive or contaminated liquids
- Wastewater and sludge pumping
- Low-maintenance applications

(Karassik et al., 2008)

Limitations

- Low efficiency compared to other pumps
- Requires continuous compressed air supply
- Not suitable for high discharge or high head
- Poor control over discharge rate

(Fox et al., 2015)

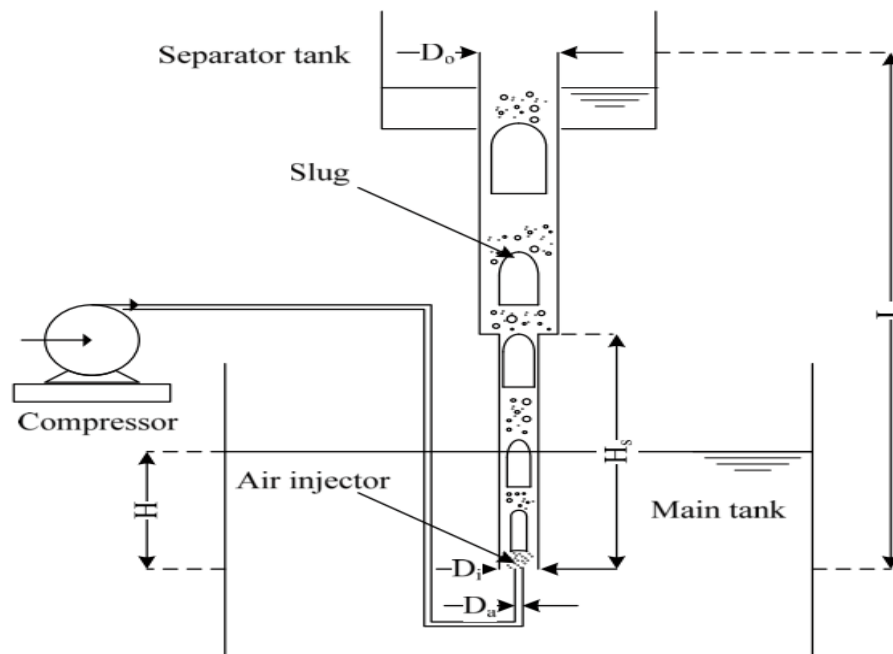


Fig. 1. Schematic of airlift pump with step geometry. (Hanafizadeh et al., 2010)

Schematic Explanation

- **Air compressor** → supplies compressed air
- **Injector nozzle** → introduces air into riser pipe
- **Riser pipe** → carries air–water mixture upward
- **Discharge outlet** → releases lifted water

(White, 2016)

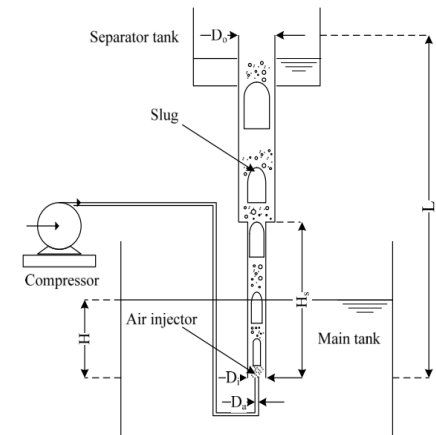
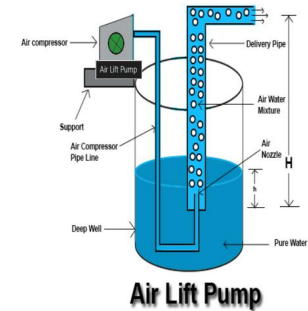


Fig. 1. Schematic of airlift pump with step geometry.

(Hanafizadeh et al., 2010)

Working principle



- **Air injection:-** Compressed air is injected near the bottom of a vertical riser pipe that is submerged in water.
- **Formation of air–water mixture:-** The injected air forms bubbles and mixes with water inside the pipe, creating a two-phase (air + water) mixture.
- **Reduction in density:-** The average density of the air–water mixture inside the riser pipe becomes lower than the density of the surrounding water outside the pipe.
- **Buoyant force and upward flow:-** Due to this density difference, a buoyant force acts on the lighter mixture, causing it to rise upward through the pipe.
- **Discharge at the outlet:-** As the mixture reaches the top, air escapes to the atmosphere and water is discharged continuously.

(Karassik et al., 2008, Fan et al., 2013)

Conclusion

- Air-lift pump works on the density difference principle
- Best suited for deep, sandy, and contaminated water
- Low efficiency but high reliability and durability

References

- Abou Taleb, F. S., & Al-jarrah, J. A. (2017). Experimental Study of an Air Lift Pump. *Engineering, Technology & Applied Science Research*, 7(3), 1676–1680. <https://doi.org/10.48084/etasr.1207>
- Di Gerlando, A., Foglia, G. M., Iacchetti, M., Perini, R., Gatelli, F., & Armellin, L. (2010). Torque and losses in the metallic shell of a dual mechanical port PM electrical machine, for dangerous fluid pumps. *19th International Conference on Electrical Machines, ICEM 2010*. <https://doi.org/10.1109/ICELMACH.2010.5607880>
- Fan, W., Chen, J., Pan, Y., Huang, H., Arthur Chen, C. T., & Chen, Y. (2013). Experimental study on the performance of an air-lift pump for artificial upwelling. *Ocean Engineering*, 59, 47–57. <https://doi.org/10.1016/j.oceaneng.2012.11.014>
- Hanafizadeh, P., Karimi, A., & Saidi, M. H. (2010). *Effect of Step Geometry on the Performance of the Airlift Pump* †.
- Mahrous, A. F. (2013). Experimental Study of Airlift Pump Performance with S-Shaped Riser Tube Bend. *International Journal of Engineering and Manufacturing*, 3(1), 1–12. <https://doi.org/10.5815/ijem.2013.01.01>
- Pougatch, K., & Salcudean, M. (2008). Numerical modelling of deep sea air-lift. *Ocean Engineering*, 35(11–12), 1173–1182. <https://doi.org/10.1016/j.oceaneng.2008.04.006>
- Zimmerman, W. B., Hewakandamby, B. N., Tesař, V., Bandulasena, H. C. H., & Omotowa, O. A. (2009). On the design and simulation of an airlift loop bioreactor with microbubble generation by fluidic oscillation. *Food and Bioproducts Processing*, 87(3), 215–227. <https://doi.org/10.1016/j.fbp.2009.03.006>
- Fox, R. W., McDonald, A. T., & Pritchard, P. J. (2015). *Introduction to fluid mechanics* (8th ed.). John Wiley & Sons.
- Karassik, I. J., Messina, J. P., Cooper, P., & Heald, C. C. (2008). *Pump handbook* (4th ed.). McGraw-Hill.
- White, F. M. (2016). *Fluid mechanics* (8th ed.). McGraw-Hill Education.

Centrifugal Pump

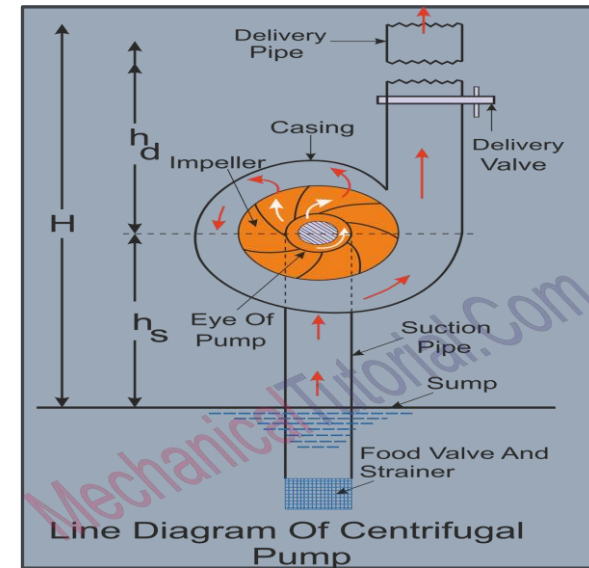
Presented by
Roshan Karki
078BCE129

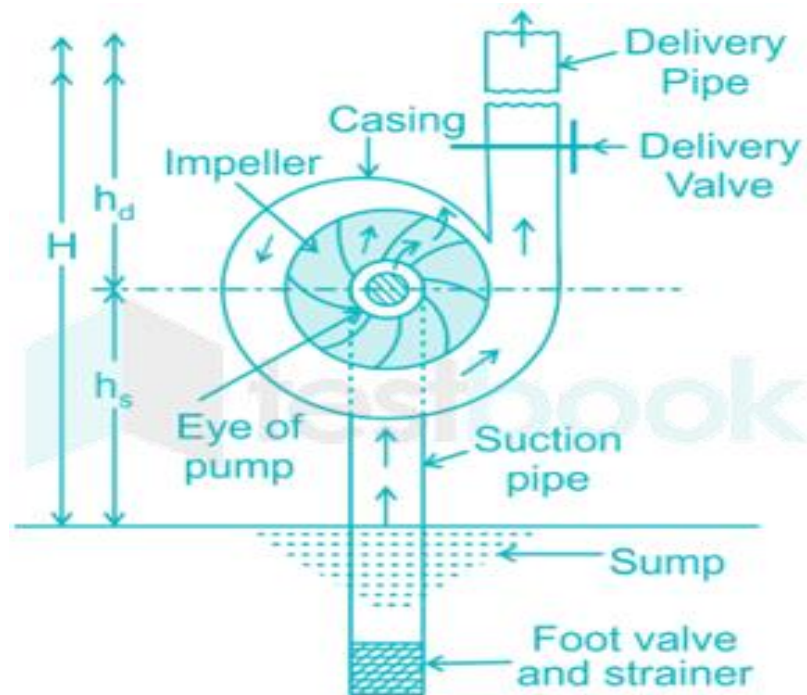
Definition

- A **centrifugal pump** is a mechanical device used to transport liquids.
- It works by converting **rotational energy** from a motor into **kinetic and pressure energy** of the fluid.
- The liquid enters the pump at the center of the **impeller** and is thrown outward by **centrifugal force**, increasing its velocity and pressure.

Working principle

- Works on the principle of centrifugal force
- Pump casing and suction pipe are primed with liquid
- Impeller rotates when motor starts
- Liquid enters at the eye (center) of the impeller
- Liquid is thrown outward by centrifugal action
- Velocity energy is converted into pressure energy in casing
- High-pressure liquid is discharged through delivery pipe





Main parts of a centrifugal pump

Suitability of Centrifugal Pump

- Pumping large quantities of liquid
- Applications requiring **continuous and uniform flow**
- Situations where simple design and low maintenance are needed
- Low to medium head operations
- Irrigation and drainage
- Clean or slightly contaminated fluids
- Fire-fighting systems

Limitations

- Not suitable for very high head
- Poor performance with high-viscosity fluids
- Efficiency drops at low discharge
- Cannot handle large solid particles
- Cavitation

Factors to be Considered in the Selection of Pump Sets

Presented by: Roshan Bahadur Bhandari (078BCE130)

□ Introduction

- Pump selection is a critical aspect of groundwater development.
- Proper selection ensures:
 - I. Efficient lifting of groundwater
 - II. Economical operation
 - III. Long service life of pump and well

(Raghunath, 2006)

❑ Need for Proper Pump Selection

➤ Incorrect pump selection may lead to:

I. Excessive power consumption

II. Low discharge

III. Damage to well and pump

IV. Uneconomical operation

(Raghunath, 2006)

❑ Factors Affecting Selection of Pump Sets

- Required discharge
- Total head to be developed
- Characteristics of aquifer and well
- Type of well
- Power availability
- Efficiency of pump
- Economic considerations

□ Required Discharge

- Discharge depends on:
 - I. Water demand (irrigation, domestic, industrial)
 - II. Safe yield of aquifer
 - III. Pumping test results
- The selected pump discharge must not exceed the safe yield of the well, otherwise:
 - I. Excessive drawdown occurs
 - II. Well efficiency reduces
 - III. Aquifer depletion may happen
- Therefore, pump capacity should be slightly less than or equal to the well yield.

(Bear, 1979)

□ Total Head to be Developed

- The pump must be capable of developing sufficient head to lift water from the pumping water level to the delivery point (Raghunath, 2006).
- Total head includes:
 - I. Static water level: Vertical distance from ground level to static water level
 - II. Drawdown in the well: Additional lowering of water level during pumping
 - III. Friction losses in pipes: Losses due to pipes, bends, valves, etc.
 - IV. Delivery head: Height from ground level to the point of water discharge
(Todd & Mays, 2005)

□ Aquifer and Well Characteristics

- Depth of aquifer: Deep aquifers require deep-well pumps such as submersible or turbine pumps, while shallow aquifers can be served by centrifugal pumps (Raghunath, 2006)
- Type of aquifer: Unconfined(Jet/Centrifugal), Confined(Deep-well turbine, Submersible)
- Well diameter and depth: Large diameter wells can accommodate larger pumps
- Yield of the well: Low-yield wells require low discharge pumps

(Bear, 1979)

□ Type of Well

- The pump must be physically and hydraulically compatible with the well.
- Shallow wells: Centrifugal pumps
- Deep tube wells: Submersible pumps, Vertical turbine pumps
- Dug wells: Jet pumps

(Todd & Mays, 2005)

❑ Power Availability

- Selection depends on available power source; electric motor or diesel engine.
- Availability and reliability of electricity
- Cost of fuel or electricity
- Maintenance requirement

(Raghunath, 2006)

❑ Economic Considerations

- The most suitable pump is not the cheapest, but the one with minimum overall cost.
- Initial cost of pump and motor
- Installation cost
- Operating and maintenance cost
- Life cycle cost

(Raghunath, 2006)

□ Conclusion

- The selection of pump sets for groundwater lifting depends on a combination of; hydraulic factors, well and aquifer characteristics, power availability, efficiency and economics.
- A properly selected pump ensures:
 - I. Sustainable groundwater extraction
 - II. Efficient operation
 - III. Longer life of well and pump
 - IV. Reduced energy and maintenance cost

(Bear, 1979; Todd & Mays, 2005)

□ References

- Bear, J. (1979). *Hydraulics of groundwater*. New York, NY: McGraw-Hill.
- Raghunath, H. M. (2006). *Groundwater* (2nd ed.). New Delhi, India: New Age International Publishers.
- Todd, D. K., & Mays, L. W. (2005). *Groundwater hydrology* (3rd ed.). Hoboken, NJ: John Wiley & Sons.